MISB MOTION IMAGERY STANDARDS BOARD	MISB ST 1107.4
STANDARD	
Metric Geopositioning Metadata	**21 October 2021**

1 Scope

MISB ST 1107 defines mandatory and objective metadata items for photogrammetric applications. This standard defines a Local Set (LS) with select metadata items from MISB ST 0801 [1], MISB ST 1010 [2], and MISB ST 1202 [3]. The metadata items specific to metric sensing are a subset of ST 0801 photogrammetric metadata items.

Note: Systems developed to read and process version 3 of ST 1107 (i.e., ST 1107.3) may not be able to read and process version 4 of ST 1107 (i.e., ST 1107.4) data due to the use of Report-on-Change in version 4. See Section 8.1 for more information.

2 References

MISB references cited herein reflect versions current to the publication date of a document. In the event of a MISB document correction, the corrected document will have a single letter Minor Version appended to the Major Version number per the MISB Document Development Process [4]. For example, corrections to ST 0001.2, which has a Major Version of 2, result in a new version titled ST 0001.2a, which includes a Minor Version of "a". The MISB will not update the referring document (this document) with the Minor Version number change. When acquiring any MISB reference listed below from an NGA repository the latest version may be either a Major or Minor Version.

[1] MISB ST 0801.9 Photogrammetry Metadata for Motion Imagery, Oct 2021.

[2] MISB ST 1010.3 Generalized Standard Deviation and Correlation Coefficient Metadata, Oct 2016.

[3] MISB ST 1202.2 Generalized Transformation Parameters, Feb 2015.

[4] MISB Document Development Process, Feb 2020.

[5] MISB ST 0807.25 MISB KLV Metadata Registry, Jun 2020.

[6] MISB ST 0107.5 KLV Metadata in Motion Imagery, Oct 2021.

[7] MISB ST 1201.5 Floating Point to Integer Mapping, Jun 2021.

[8] ISO/IEC 8825-1:2021 Information Technology – ASN.1 encoding rules: Specification of Basic Encoding Rules (BER), Canonical Encoding Rules (CER) and Distinguished Encoding Rules (DER).

[9] MISB MISP-2022.1 Motion Imagery Standards Profile, Oct 2021.

[10] MISB ST 0603.5 MISP Time System and Timestamps, Oct 2017.

[11] MISB MISP-2022.1: Motion Imagery Handbook, Oct 2021.

3 Acronyms and Definitions

BER-OID	Basic Encoding Rules – Object Identifier
CE90	Circular Error at 90% Confidence
CRC	Cyclic Redundancy Check
DGMS	Direct Geopositioning Metric Sensor
FFOV	Full Field-of-View
FLP	Floating Length Pack
GPS	Global Positioning System
IMU	Inertial Measurement Unit
ISR	Intelligence, Surveillance and Reconnaissance
KLV	Key-Length-Value
LE	Linear Error
LRF	Laser Range Finder
LS	Local Set
MISB	Motion Imagery Standards Board
MISP	Motion Imagery Standards Profile
PED	Processing, Exploitation, and Dissemination
ROC	Report-on-Change
SACP	Single Aim-point Center Pixel
SDCC	Standard Deviation and Correlation Coefficient
ST	Standard
TLE	Target Location Error
ZLI	Zero-Length-Item

4 Revision History

Revision	Date	Summary of Changes
ST 1107.4	10/21/2021	• Added Report-on-Change (RoC) ○ Systems developed to read and process version 3 of ST 1107 (i.e., ST 1107.3) may not be able to read and process version 4 of ST 1107 (i.e., ST 1107.4) data due to the use of Report-on-Change in version 4. See Section 8.1 for more information. • Editorial updates • Added Reqs -13, -14, -15, -16, -17 & -18 • Changed document title • Updated References [1], [4], [5], [6], [7], [9]
	05/17/2022	• Added Generalized Transformation LS item to the list of items that cannot use a ZLI.

5 Introduction

A metric sensor collects sufficient metadata to support the computation of a target coordinate (latitude, longitude, and height-above-ellipsoid), and the coordinate's uncertainty, known as Target Location Error (TLE) or Circular and Linear Error (CE/LE). A Direct Geopositioning Metric Sensor (DGMS) enables the computation of the target coordinate(s) and uncertainties from a single image by integrating a Laser Range Finder (LRF) (Single Aim Point Center Pixel – SACP) or a framing LIDAR sensor (Full Field of View – FFOV) into the sensor system. The value of a DGMS is the ability to generate target coordinates and an error estimate (TLE or CE/LE) for those coordinates with a known level of confidence resulting from direct calculation.

To exploit a metric sensor or a DGMS requires two critical elements: (1) a rigorous sensor model; and (2) a complete set of metadata describing the sensor state and the measurement uncertainties of its state. These elements enable a myriad of down-stream Processing, Exploitation, and Dissemination (PED) capabilities, such as allowing applications to combine imagery with other imagery or data sources (i.e., data fusion). The GWG/Community Sensor Model Working Group manages the sensor model; however, this standard defines the metadata items to describe the sensor state and the measurement uncertainties.

Integrating metric capability with Motion Imagery is increasingly important as Motion Imagery plays a more significant role in fulfilling ISR mission needs. The photogrammetric metadata defined in MISB ST 0801 provides the required items to describe a sensor with sufficient content to compute precision geolocations. MISB ST 1010 conveys the variance-covariance information about the parameters in ST 0801. The first 31 items, and Item 47, of the LS are the items in ST 0801 having uncertainty information (consistent with the order required in ST 1010). The ST 1010 Standard Deviation and Correlation Coefficient Floating Length Pack (FLP) for these items immediately follow the values in the LS. The remaining items of the LS are items in ST 0801 which do not have an uncertainty model.

6 Accuracy and Metricity

The terms "accuracy" and "metricity" have two different but related definitions. Accuracy is a measure of how well a system can calculate the location of a point of interest compared to its actual location in the real world. A more accurate sensor can produce target coordinates closer to the true location of a coordinate (i.e., the missed distance is small) than a less accurate sensor. System requirements usually state accuracy needs, which is dependent on how well a system measures its state when collecting an image. A system may improve its accuracy by using higher quality system components (e.g., improved Inertial Measurement Unit (IMU) or Global Positioning System (GPS) solution). Understanding the accuracy of a sensor's metadata requires the measurement of uncertainties (errors); this refers to the metricity.

Metricity provides confidence in the calculated location of a point of interest. Systems express the confidence in terms of uncertainty predictions for various components of the geopositioning result; therefore, confidence depends on how well the system knows the uncertainties (errors) associated with the measured system parameters for each image. As a metric sensor is capturing an image, the metric sensor reports the dynamic system information as metadata items. Even when values have large uncertainties and inaccurate data, the sensor is metric. A system which does not provide current error estimates for dynamic system values is not a metric sensor.

Figure 1 illustrates this relationship between accuracy and metricity. The lower-left quadrant represents a less accurate, non-metric system. The calculated target location shows a large displacement when compared to the actual geolocation of the target. By improving the system components, the system may become more accurate and move into the lower-right quadrant. For both non-metric cases, the confidence in the calculated target location is unknown.

If, however, the less accurate, non-metric system of the lower-left quadrant provided error estimates (e.g., CE90) for the dynamic system parameters, it becomes a metric sensor and moves to the upper-left quadrant. While such a system may not improve in accuracy, the confidence in the calculated target location is known and users may use it for engagement, collateral damage assessment, weapons effect calculations or other precision-based tasks. The ideal case is where the system components are of sufficient high quality for accuracy and produce error estimates (metric) for the dynamic system parameters. This is the case shown in the upper-right quadrant, and such a system can provide even better actionable target information.

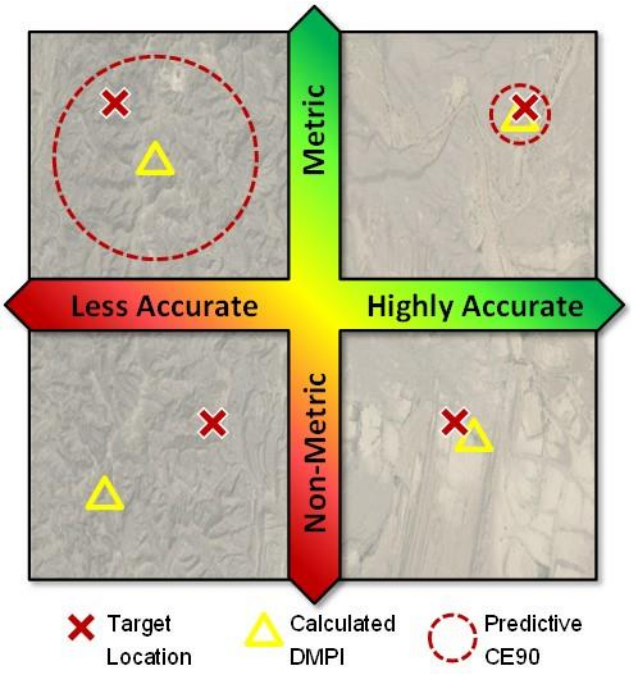


Figure 1: Relationship between Accuracy and Metricity

7 Metadata Timing

The design and understanding of system timing are extremely important in a metric sensor system. The Metric Geopositioning LS includes a metadata item to record the time for when a set of metadata items are valid. Uncertainties and misalignments in timing can introduce additional error in the uncertainty of calculated target coordinates. Systems which implement this standard should have the capability to capture and timestamp both the Motion Imagery and the metadata at the same instant in time. The uncertainty (error) estimates must account for any timing differences between the metadata items themselves, or between the metadata items and the captured image.

8 Metric Geopositioning Local Set (LS)

The Metric Geopositioning Local Set contains metadata items for photogrammetric applications. The Metric Geopositioning Local Set 16-Byte Universal Label registered in MISB ST 0807 [5] is:

06.0E.2B.34.02.0B.01.01.0E.01.03.03.22.00.00.00 (CRC 13780)

8.1 Report-on-Change

MISB ST 1107 assumes the Report-on-Change (ROC) system described in the Motion Imagery Handbook. With exception of the *MANDATORY* items in Table 1, items update when their value changes or if a Metadata Refresh Period (i.e., 30-second period) has elapsed since the last item update. Receivers treat an item as *Unknown* when the item does **not** update within a Metadata Refresh Period.

MISB ST 0107 [6] provides a set of baseline requirements when implementing Report-on-Change.

Requirement	
ST 1107.4-13	Metric Geopositioning LS items implemented using Report-on-Change shall adhere to the requirements for usage stated in MISB ST 0107.

All items within the Metric Geopositioning LS may use a Zero-Length-Item (ZLI), except the following: items designated as MANDATORY, MANDATORY-ROC, and MANDATORY-ALT and the Generalized Transformation LS (Item 33).

8.2 Local Set Items

Table 1 lists the Metric Geopositioning LS items. Many of the items in Table 1 refer to other documents which contain additional details regarding the data type, size, and the mappings for integers, if applicable. Encode the metadata items originating from other standards according to the definition, rules, and requirements within those documented standards.

Table 1 has five columns: Tag, Name, Type, Directive, and Uncertainty Data.

- *Tag* indicates the Tag specifying the metadata item
- *Name* is the label associated with the Tag
- *Type* describes the method of encoding the value in binary; the allowed values are:
  - IMAPB indicates the use of MISB ST 1201 [7] to encode the floating-point value; refer to MISB ST 0801 for details on the minimum and maximum ranges for the mapping of the specific value.
  - float indicates the use of either 16, 32, or 64-bit IEEE Floating Point values
  - uint indicates an unsigned integer
  - LS indicates the item value contains a KLV Local Set
  - FLP indicates the item value contains a Floating Length Pack
  - BER-OID indicates the encoding of an item is a BER OID value (ISO 8825-1 [8])

The *Directive* column indicates the item as one of: Mandatory, Mandatory-ROC, Mandatory-ALT or Objective.

- Mandatory items must be in every MISB ST 1107 packet
- Mandatory-ROC items must follow the ROC rules
- Mandatory-ALT items must follow the rules in the item's description
- Objective items are optional and may use ROC as needed

Together, ***Mandatory***, ***Mandatory-ROC***, and ***Mandatory-ALT*** items represent core items required for data exploitation. The additional ***Objective*** items complete an ideal set of items for a DGMS, which may yield results with the highest fidelity. Computing Single Aim-point Center Pixel (SACP) or Full Field of View (FFOV) exploitation requires the Objective items.

The *Uncertainty Data* column defines the ST 1010 Source List for the SDCC-FLP in Item 32 along with the type of value. There are three values in the Uncertainty Data column:

- N/A indicates Not Applicable (i.e., not included in the Source List)
- IMAPB indicates the use of MISB ST 1201 to encode the floating-point value using the given minimum and maximum values, in the form IMAPB(Min, Max)
- float indicates the use of either 16, 32 or 64-bit IEEE floating point values

The order of the item instances in the LS is important when building the value for Item 32. At run-time, systems determine the length of each value in the Local Set based on the accuracy requirements and bandwidth availability.

Table 1: Metric Geopositioning LS

Tag	Name	Type	Directive	Uncertainty Data
1	Sensor ECEF Position Component X	IMAPB[1]	**MANDATORY**	IMAPB(0,650)
2	Sensor ECEF Position Component Y	IMAPB[1]	**MANDATORY**	IMAPB(0,650)
3	Sensor ECEF Position Component Z	IMAPB[1]	**MANDATORY**	IMAPB(0,650)
4	Sensor ECEF Velocity Component X	IMAPB[1]	OBJECTIVE	IMAPB(0,70)
5	Sensor ECEF Velocity Component Y	IMAPB[1]	OBJECTIVE	IMAPB(0,70)
6	Sensor ECEF Velocity Component Z	IMAPB[1]	OBJECTIVE	IMAPB(0,70)
7	Sensor Absolute Azimuth	IMAPB[1]	**MANDATORY**	IMAPB(0,0.2)
8	Sensor Absolute Pitch	IMAPB[1]	**MANDATORY**	IMAPB(0,0.2)
9	Sensor Absolute Roll	IMAPB[1]	**MANDATORY**	IMAPB(0,0.2)
10	Sensor Absolute Azimuth Rate	IMAPB[1]	OBJECTIVE	IMAPB(0,1)
11	Sensor Absolute Pitch Rate	IMAPB[1]	OBJECTIVE	IMAPB(0,1)
12	Sensor Absolute Roll Rate	IMAPB[1]	OBJECTIVE	IMAPB(0,1)
13	Boresight Offset Delta X	IMAPB[1]	OBJECTIVE	IMAPB(0,650)
14	Boresight Offset Delta Y	IMAPB[1]	OBJECTIVE	IMAPB(0,650)
15	Boresight Offset Delta Z	IMAPB[1]	OBJECTIVE	IMAPB(0,650)
16	Boresight Delta Angle 1	IMAPB[1]	OBJECTIVE	IMAPB(0,2)
17	Boresight Delta Angle 2	IMAPB[1]	OBJECTIVE	IMAPB(0,2)

[1] see MISB ST 0801 [1] for min and max values; systems define IMAP length as the system's precision requires.

18	Boresight Delta Angle 3	IMAPB[1]	OBJECTIVE	IMAPB(0,2)
19	Focal Plane Principal Point Offset Y	IMAPB[1]	**MANDATORY-ROC**	IMAPB(0,1)
20	Focal Plane Principal Point Offset X	IMAPB[1]	**MANDATORY-ROC**	IMAPB(0,1)
21	Sensor Calibrated / Effective Focal Length	IMAPB[1]	**MANDATORY-ALT**	IMAPB(0,350)
22	Radial Distortion Constant Parameter	float	OBJECTIVE	float
23	First Radial Distortion Parameter	float	OBJECTIVE	float
24	Second Radial Distortion Parameter	float	OBJECTIVE	float
25	Third Radial Distortion Parameter	float	OBJECTIVE	float
26	First Tangential / Decentering Parameter	float	OBJECTIVE	float
27	Second Tangential / Decentering Parameter	float	OBJECTIVE	float
28	Third Tangential / Decentering Parameter	float	OBJECTIVE	float
29	Differential Scale Affine Parameter	float	OBJECTIVE	float
30	Skewness Affine Parameter	float	OBJECTIVE	float
31	Slant Range	float	OBJECTIVE	IMAPB(0,650)
32	Standard Deviation and Correlation Coefficient FLP	FLP	**MANDATORY**	N/A
33	Generalized Transformation LS	LS	OBJECTIVE	N/A
34	Image Rows	uint	**MANDATORY-ROC**	N/A
35	Image Columns	uint	**MANDATORY-ROC**	N/A
36	Pixel Size X	IMAPB[1]	**MANDATORY-ROC**	N/A
37	Pixel Size Y	IMAPB[1]	**MANDATORY-ROC**	N/A
38	Slant Range Pedigree	uint	OBJECTIVE	N/A
39	Line Coordinate	float	OBJECTIVE	N/A
40	Sample Coordinate	float	OBJECTIVE	N/A
41	LRF Divergence	float	OBJECTIVE	N/A
42	Valid Range of Radial Distortion	float	OBJECTIVE	N/A
43	Precision Time Stamp	uint64	**MANDATORY**	N/A
44	Document Version	BER-OID	**MANDATORY**	N/A
45	CRC-16-CCITT	uint16	**MANDATORY**	N/A
46	Leap seconds	int	OBJECTIVE	N/A
47	Sensor Calibrated / Effective Focal Length Extended	IMAPB[1]	**MANDATORY-ALT**	IMAPB(0,350)

9 Metric Geopositioning LS Item Details

Requirement(s)	
ST 1107.1-01	All metadata shall be expressed in accordance with MISB ST 0107.
ST 1107.1-02	All metadata items indicated as MANDATORY in MISB ST 1107 Table 1 shall be populated and transmitted in the Metric Geopositioning LS.
ST 1107.4-14	A metadata item indicated as MANDATORY-ROC in MISB ST 1107 Table 1 shall be in the Report-on-Change "Known-Changing" or "Known-Static" state.

9.1 Item 1 through Item 20 – MISB ST 0801 Defined

Refer to MISB ST 0801 for a complete description of the parameters listed for metadata Item 1 through Item 20.

9.2 Item 21 Sensor Calibrated / Effective Focal Length

Item 47 is a value range extension of Item 21. The local set retains Item 21 for backwards compatibility, therefore:

1) Systems producing Metric Geopositioning LS should use Item 47 instead of Item 21.
2) Systems reading Metric Geopositioning LS data need to be able to parse both Item 21 and Item 47. This standard requires the use of one or the other but not both values in the same metadata stream.

See Section 9.11 for the rules, utilization, and Report-on-Change requirements.

Refer to MISB ST 0801 for a complete description of the Sensor Calibrated / Effective Focal Length.

9.3 Item 22 through Item 31 – MISB ST 0801 Defined

Refer to MISB ST 0801 for a complete description of the parameters listed for metadata items Item 22 through Item 31.

9.4 Item 32 - Standard Deviation and Correlation Coefficient FLP (SDCC-FLP)

Each of the values listed for metadata items Item 1 through Item 31 and Item 47 can have uncertainty (i.e., standard deviation or sigma, σ) information. Additionally, each value can be correlated to any of the other 30 values resulting in a potential correlation coefficient value for a particular pair of values. Values with no correlation result in a correlation coefficient value of zero for that pair of values.

MISB ST 1010 defines how to package the standard deviation and correlation coefficient values into the SDCC-FLP for the measurements made in Items 1 through 31 and 47. Item 1 through Item 31 along with Item 47 define the Source List (see ST 1010) for MISB ST 1107. Per ST 1010, systems define Refined Source Lists at run-time, based on those standard deviation and correlation coefficient values available.

Per ST 1010, systems write the Refined Source List values into the Metric Geopositioning LS (in any order) immediately followed by the SDCC-FLP, where each row of the SDCC-FLP upper triangular matrix is in the same order as the values just written in the LS.

The Metric Geopositioning LS may utilize multiple SDCC-FLPs; however, the SDCC-FLP is the only item in the Metric Geopositioning LS that can have multiple uses. See ST 1010 for SDCC usage examples.

The SDCC-FLP has five defining parameters: Matrix Size, Parse Control, Bit Vector, Standard Deviation Elements (values), and the Correlation Coefficient Elements (values).

9.4.1 Matrix Size

The Matrix Size is set to the item count in the Refined Source List. This value will be less than or equal to 31, the size of the Source List.

9.4.2 Parse Control

ST 1107 requires the use of Mode 2 Parse Control. Consult MISB ST 1010 for further description of Mode 1 and 2 of the Parse Control.

Requirement	
ST 1107.2-12	The Metric Geopositioning Local Set shall only include SDCC-FLPs using Mode 2 Parse Control, as defined in MISB ST 1010.

Systems compute the five values in the Mode 2 Parse Control at runtime: C_s, S_f, S_{len}, C_f, and C_{len}.

- The C_s value indicates if the correlation coefficient values use a sparse representation in the SDCC-FLP (see Section 9.4.3).
- The S_f value defines the data format type of the standard deviation values, either IMAP (see MISB ST 1201) or IEEE Floating Point values. ST 1010 does not allow the mixing of types; therefore, systems convert all standard deviation values to one type. Systems need to determine how to balance the desired precision and bandwidth requirements. If exterior orientation parameters (Items 1 through 18 and 31, 47) are the only Refined Source List items, the recommended data type and size are two-byte IMAPB values. Table 1 defines the IMAPB upper and lower bounds for each respective item. If systems include interior orientation parameters (Items 19 through 30), or exterior and interior orientation parameters in the Refined Source List, the recommended data type for all standard deviation values is four-byte IEEE Floating Point. ST 1010 does not support mixed data types for standard deviation values, so this standard uses the most conservative value if two different data types are present.
- The S_{len} value defines the number of bytes used by each standard deviation value. If a system requires greater precision, systems may add more bytes.
- The C_f value defines the data format type of the correlation coefficient values, either IMAP (see MISB ST 1201) or IEEE Floating Point values. This standard recommends IMAP to save bandwidth. The mapping is IMAPB (-1.0, 1.0, C_{len}) for all correlation coefficient values with a recommended value for C_{len} equal to two (2) bytes. This does not limit the use of additional bytes if a system requires greater precision.
- The C_{len} value defines the number of bytes for each correlation coefficient value. Systems requiring greater precision can use more bytes.

9.4.3 Bit Vector

As discussed in MISB ST 1010, correlation coefficient data can be a sparse matrix. The Bit Vector indicates where to eliminate the zeros in the SDCC-FLP. See Appendix A in ST 1010 to determine when systems should use the Bit Vector. Systems can decide to use the Bit Vector at run time.

9.4.4 Standard Deviation Values

Systems convert standard deviation values to the desired data format (if necessary), either IMAP or IEEE Floating Point, and include in the SDCC-FLP in the same order of the Refined Source List.

9.4.5 Correlation Coefficient Values

Systems convert the correlation coefficient values to the desired data format (if necessary), either IMAP or IEEE Floating Point, and include in the SDCC-FLP. The rows and columns of the correlation coefficient matrix are in the same order as the Refined Source List.

9.5 Item 33 – Generalized Transformation LS

The Generalized Transformation LS is an optional set of data captured in Item 33, which relates the virtual image coordinate system to the distorted image coordinate system. The Generalized Transformation LS may appear up to four times in the Metric Geopositioning LS to account for all enumerations defined in MISB ST 1202.

9.6 Item 34 through Item 42 – ST 0801 Defined

Refer to MISB ST 0801 for a complete description of the parameters listed for metadata items Item 34 through Item 42.

9.7 Item 43 – Precision Time Stamp

The Precision Time Stamp is the time when the system measures the metadata in this LS. The Motion Imagery Standards Profile (MISP [9]) requires the insertion of the Precision Time Stamp into every frame of Motion Imagery; therefore, the Precision Time Stamp provides a deterministic relationship between a specific image within a sensor's imagery and the values in the LS. For most imaging systems, a Precision Time Stamp will denote a time between the start and end of the image integration period. It is important that systems register metadata to the image the metadata applies to. Synchronization of an image with its metadata is critical for photogrammetric exploitation of the data. Unknown misalignments between an image and metadata can be detrimental to precise geopositioning exploitation. Refer to MISB ST 0603 [10] for further information about timestamps.

Item 43, Precision Time Stamp, does not include leap seconds and therefore the Precision Time Stamp does not represent UTC. To convert the Precision Time Stamp to UTC, follow the instructions in Section 9.10 (Item 46 - Leap Seconds).

9.8 Item 44 – Document Version

The Document Version identifies the version of this document a system uses to implement the metadata set. For example, if a system implements ST 1107.4, the value for this Item will be "4".

9.9 Item 45 – CRC-16-CCITT

To help detect erroneous metadata post transmission, systems include a two-byte CRC in every LS as the last item. Systems compute the CRC across the entire LS, starting with the 16-byte LS key and ending with the length field of the CRC data element. Figure 2 illustrates the input data range for the checksum. If the calculated CRC of the received LS packet does not match the CRC stored in the packet, the receiver discards the packet as being invalid. Please refer to the Motion Imagery Handbook [11] for details on computing the CRC value.

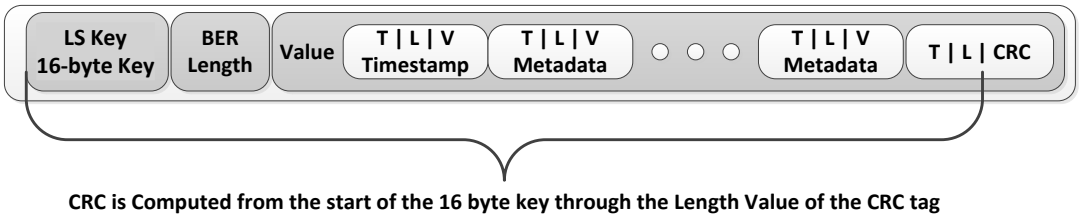


Figure 2: CRC Representation

9.10 Item 46 – Leap Seconds

The Leap Seconds item is the positive or negative count of leap seconds for the given time. To convert the Precision Time Stamp value to UTC, multiply the leap second value by 1,000,000 to convert it to microseconds, then add it to the Precision Time Stamp. See the Motion Imagery Handbook for more details on Leap Seconds and the MISP Time System.

9.11 Item 47 – Sensor Calibrated / Effective Focal Length Extended

Refer to MISB ST 0801 for a complete description of the Sensor Calibrated / Effective Focal Length Extended.

Item 47 is a value range extension of Item 21. The local set retains Item 21 for backwards compatibility, therefore:

1) Systems producing Metric Geopositioning LS should use Item 47 instead of Item 21.
2) Systems reading Metric Geopositioning LS data need to parse both Item 21 and Item 47, but this standard requires the use of one but not both values.

A series of Metric Geopositioning LS packets forms a Metric Geopositioning stream (see Motion Imagery Handbook). The Metric Geopositioning stream requires including one of the *Sensor Calibrated / Effective Focal Length* values, either Item 21 or Item 47, but not both. When using either item, the item may use Report-on-Change, but the item must always have a value (i.e., the item cannot be in the Unknown Report-on-Change state).

Requirement(s)	
ST 1107.4-15	Where the Metric Geopositioning Local Set stream includes Item 21 in either the "Known-Changing" or "Known-Static" Report-On-Change state, the Report-On-Change state of Item 47 shall be "Unknown".
ST 1107.4-16	Where the Metric Geopositioning Local Set stream includes Item 47 in either the "Known-Changing" or "Known-Static" Report-On-Change state, the Report-On-Change state of Item 21 shall be "Unknown".
ST 1107.4-17	Where the Metric Geopositioning Local Set stream includes Item 21, Item 21 shall be in the Report-on-Change state of "Known-Changing" or "Known-Static".
ST 1107.4-18	Where the Metric Geopositioning Local Set stream includes Item 47, Item 47 shall be in the Report-on-Change state of "Known-Changing" or "Known-Static".

10 Deprecated Requirements

Requirement(s)	
ST 1107.1-03 (Deprecated)	The program office shall select from the "Objective" elements in MISB ST 1107 Table 1 to produce a data population plan that enables the full capability for their system.
ST 1107.1-04 (Deprecated)	When transmitting a Metric Geopositioning LS either the airborne platform elements or the spaceborne platform elements shall be used, but not both.
ST 1107.1-05 (Deprecated)	When the Metric Geopositioning LS is used for airborne DGMS application, real-time position ECEF values as represented by LS Tags 1, 2 and 3 shall be present.
ST 1107.1-06 (Deprecated)	When the Metric Geopositioning LS is used for spaceborne DGMS application, real-time ECEF values as represented by LS Tags 7, 8 and 9 shall be present.
ST 1107.1-07 (Deprecated)	Only one value of position information shall be transmitted in the stream.
ST 1107.1-08 (Deprecated)	Position information shall be transmitted only once per stream.
ST 1107.1-09 (Deprecated)	Only one value of velocity information shall be transmitted in the stream.
ST 1107.1-10 (Deprecated)	Velocity information shall be transmitted only once per stream.
ST 1107.1-11 (Deprecated)	Standard Deviation and Correlation Coefficient metricity information of a data element shall be conveyed in accordance with MISB ST 1010.